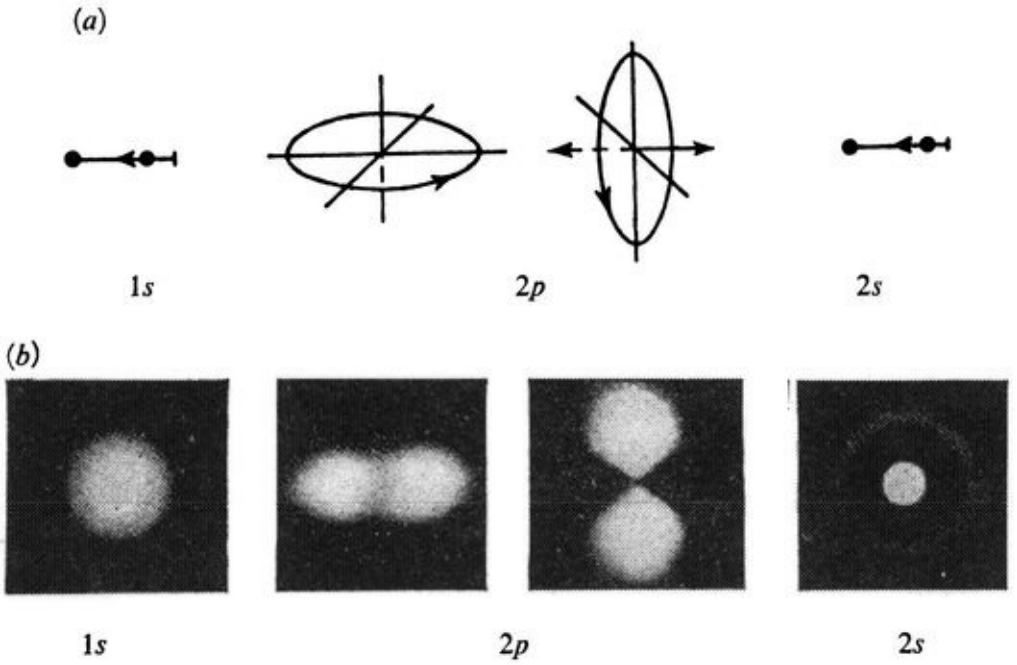
probability distribution, such as is shown in Figure 7 (*b*). It is computed from Schrödinger's wave mechanics.

But there exist also 'overtones'—excited states—through which an atom can pass as a result of the impact of a foreign electron. In such an excited state, the atom becomes capable of radiation, of emitting a *light photon* (quantum of energy of light). This happens because the atom passes from its excited state back to its ground state, or into another excited state of smaller energy. Figure 7 (*a*) and (*b*) show such excited states, marked by the symbols 2*p* and 2s. In the wave aspect of matter, the excited state closest to the ground state is represented by a stationary wave with a nodal plane perpendicular to the plane of our drawing. But this density distribution is, again, just a model, a mere aid to our visual perception, and acquires a concrete significance solely by virtue of thousands of snapshots. If we want to depict the same stationary states under the particle aspect of matter, we obtain the picture of a circular orbit, as in Bohr's original theory. But this orbit can assume many different spatial positions, and the superposition of all these possible states results in a probability or density distribution, in which the same nodal plane appears which is present in the wave aspect.




Figures 7—Hydrogen atom in the ground state and in states of excitation.

For excited states of higher energy (2*s*), a high degree of density is obtained in the centre, with a sparsely occupied ring on the outside. In such cases, there is always a certain probability that the electron may suddenly be encountered outside, in the vicinity of the ring.

We shall not go into any further details here. My sole purpose has been to